SID DIAMOND

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Personal statement

Master's year Physics student at Imperial College London researching the complexity economics of patent networks within the Theoretical Physics Group. Currently supplementing this thesis with courses in information theory and a sociological analysis examining how the use of benchmark datasets in supervised machine learning can create bias in hate speech detection systems, to inform policy. My undergraduate background includes statistical mechanics, data analysis and programming courses, a bachelor's thesis on uncertainty quantification in large language models, and the cohort's top mark in Communicating Physics for a policy-focused educational report. Last summer I undertook an internship as the lead developer for a minimal viable product multi-agent planning application at a software company. The previous summer I conducted an independent research project under the guidance of the Imperial High Performance Computing (HPC) research group using Python-based natural language processing (NLP) tools to produce a per document probability of actionable insight for economic climate policy documents.

Education

Imperial College London: Integrated MSci in Physics — 2022- Present

- Year One: 2:1| Year Two: 2:1 | Year Three: 2:1 | Masters Year: On track for 1st class
- First-class results in all programming. Object-Oriented Programming (98.7%)
- Statistics (72.9%)
- BSc Thesis in Uncertainty in Large Language models (68.50%)
- Laboratory and Problem solving (79.3%)
- Communicating Physics (84.1%)
- Statistical Mechanics (67%)
- Currently completing a master's thesis on modelling innovation networks and applying citation network analysis to the artificial networks (1st class for literature review).
- Currently also studying Information theory, A Sociology essay elective and General Relativity

Stoke Newington School and Sixth Form — 2014 -2021

- A-level Subjects: Further Maths (A*), Maths (A*), Physics (A*), Chemistry (A*), EPQ (A*)
- Chemistry Olympiad Silver 2021
- Kelvin Prize winner 2020 (*Peterhouse College Cambridge University Science Competition*)
- GCSE Subjects: 3 grade 9s, 3 grade 8s, 3 grade 7s

Research Projects

Citation Networks & Innovation | *Master's Thesis* — *(Ongoing)*

- Developing new analytical, numerical, and goodness-of-fit measures for innovation processes to generate synthetic patent citation networks modelled as directed acyclic graphs (DAGs)

- Using the artificial network results to quantitively inform on how breakthrough innovations emerge in real patent citation data (NBER Patent Citation Data File)
- Provisional first-class mark for literature review; working under the supervision of Associate Professor Tim Evans, Theoretical Physics Group, Centre for Complexity Science

BERT Economic Policy Inference Software | *Imperial HPC + BSc Thesis* Summer–Winter 2024

- Developed a Bidirectional Encoder Representations from Transformers (BERT)-based NLP tool to extract policy topics and political biases in documents, producing, for example, the per document probability of actionable insight for economic climate policy documents, achieving near-industry accuracy (ClimateBERT dataset)
- Developed into undergraduate thesis on uncertainty quantification in large language models; investigated inverse Fisher Information based variance estimates, implemented Monte Carlo dropout for transformer uncertainty, and analysed information loss from UMAP dimensionality reduction
- Worked independently over the summer under guidance from the HPC Research Group, before continuing under the supervision of Dr Aidan Crilly, Physics, Imperial College Research Fellow

A-Level Physics Attainment Gap Report | *Communicating Physics* Spring 2023

- Authored evidence-based report on state vs. private school preparation, integrating quantitative attainment data with qualitative classroom observations
- Awarded cohort-best mark; demonstrated ability to write clear, transformative insights for policymakers

Higgs Boson Data Analysis Project | *Coursework* Spring 2022

- Simulated Higgs discovery using chi-squared minimisation and hypothesis testing; demonstrated rigorous application of statistical modelling techniques

Employment

Product Engineer Intern at Mindlace Ltd Summer 2025

- Led development of an AI agent-driven planning application MVP. Built multi-agent architectures that autonomously coordinated complex booking workflows
- Used modern AI agent software (n8n, Supabase, OpenRouter, Pinceone) to create the entirely unsupervised booking of special requirement meeting itineraries requested by users

Science, English, and Maths Teacher and Tutor 2021 – 2024

- Tailored explanations to meet diverse learner needs, effectively supported a range of abilities and neurodivergence in pupils
- Demonstrated leadership in 4-month weekly placement independently teaching GCSE and A-level Physics to classrooms of 30 students at Brentside High School, Ealing

Landways Management Summer 2021

- Evidenced reconfiguration of weak network strengths in a football stadium and subsequent maintenance of the improved systems
- Installation of hardware based on technical understanding

Ove Arup Engineering Internship Spring 2019

- Implemented creative solutions to reduce project timelines on internal software

- Punctual and professional temperament

Technical Skills

- Python. Relevant libraries: PyTorch, SciPy, NumPy, Pandas, Matplotlib, Plotly
- NetLogo
- Minimal Viable Product Software Engineering tools: n8n, Shadcn/ui, Supabase, OpenRouter, Pinecone, Firecrawl, Tavily

Hobbies & Interests

Football & Running

- Imperial Men's Football 1st and 2nd teams, previously acting captain of 2nd team
- Enjoy running. Previously competitively at nation level however now recreationally

Journalism & Music

- Comment Piece writer for, Felix, the Imperial College student newspaper. Writing on: Environmental and Social Governance (ESG) and artificial intelligence
- To wind down, I like to play the guitar